Neoplasia

Nomenclature (LG8)

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Learning outlines:

1. Terminology and definition
2. Components of neoplasm
3. Classification
4. Nomenclature of benign tumors
5. Nomenclature of malignant tumors

- Neoplasia is a genetic disorder of cell growth.
- Tumors arise from clonal expansion of a transformed cell.
- Names are based mainly on cell/tissue of origin and biologic behavior.

Core terminology

Essential terms

- **Neoplasia** =means new growth
- New growth formed is known as a **neoplasm** (Greek, neo = new + plasia = growth).
- Term "**tumor**" was originally used for the swelling caused by inflammation, but it is now used synonymously with neoplasm.
- **Oncology** (Greek, oncos = tumor) is the study of tumors or neoplasms.
- **Cancer**= malignant tumors (crab)

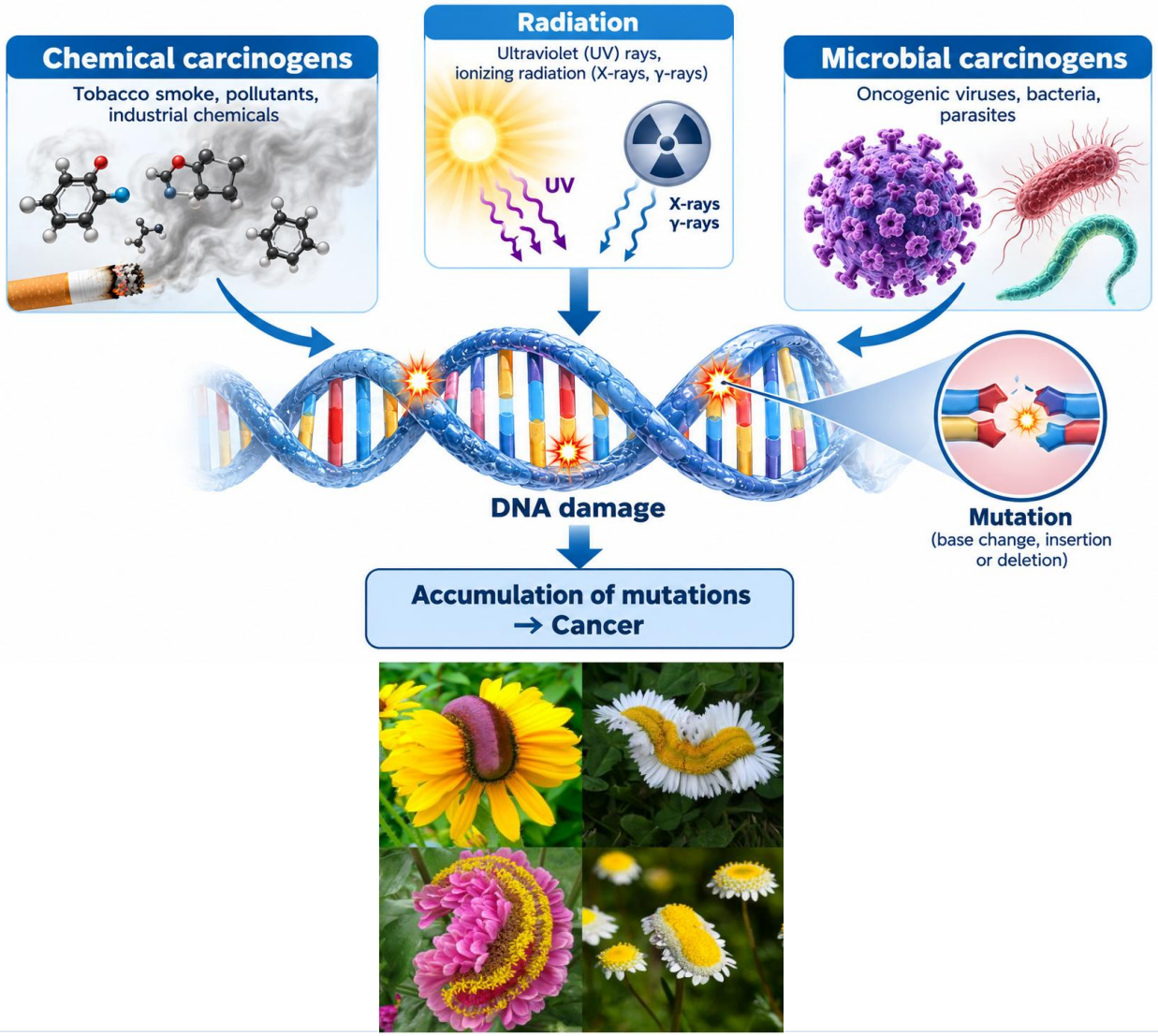
Neoplasia definition:

A neoplasm is defined as a **genetic disorder** of cell growth "**disorder of cell growth**" that is triggered by **mutations** affecting a single cell and its clonal progeny.

Willis definition:

- An abnormal mass of tissue.
- Growth is excessive and uncoordinated with normal tissue.
- Growth persists after the stimulus has ceased.

How a normal cell becomes neoplastic

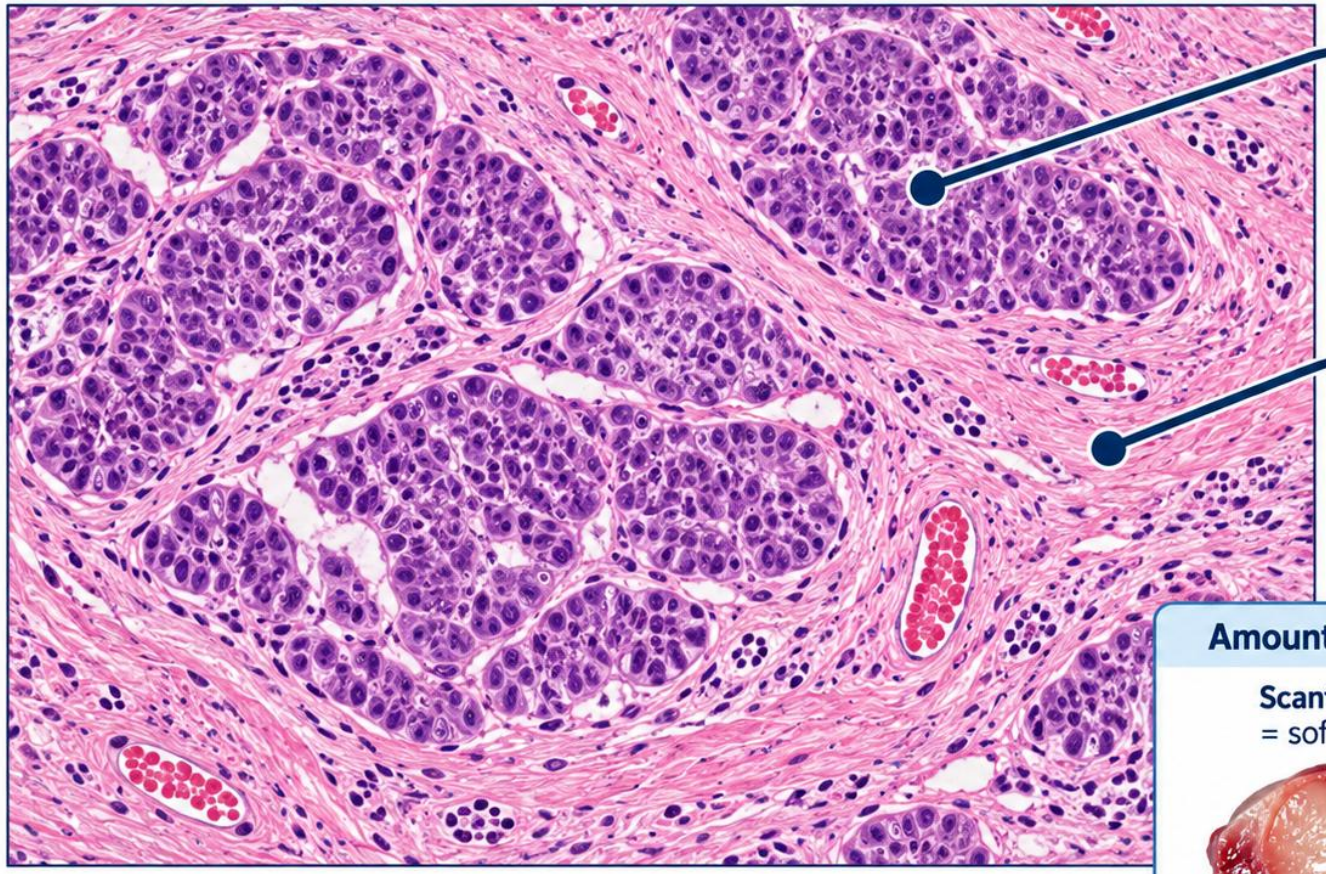


Non-lethal genetic damage in the genetic apparatus (MUTATION)

- Inherited in the germ line.
- Acquired in somatic cells.
- Major causes include chemicals, radiation and microbial.
- Mutation changes growth-control genes and produces a transformed clone.

Microscopic components of a neoplasm

Components of a Neoplasm



Parenchyma

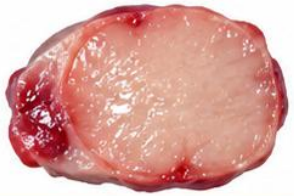
Proliferating neoplastic cells; determines tumor behavior and nomenclature.

Stroma

Connective tissue, blood vessels, inflammatory cells; supports growth of the neoplasm.

Amount of stroma affects gross consistency

Scant stroma
= soft, fleshy



Abundant desmoplastic stroma
= firm, scirrhous



CLASSIFICATION Of Neoplasia

Tumors are classified according to their biological behavior in to:

Benign tumors.

Malignant tumors.

Nomenclature Of Benign Neoplasms:

Generally named by attaching the suffix "**oma**" to the cell of origin.

1. Mesenchymal Tumors-nomenclature

Cell of origin	*Benign*
Fibrous tissue	Fibroma
Fat cell	Lipoma
Blood vessel	Hemangioma
Cartilage	Chondroma
Bone	Osteoma
Smooth muscle	Leiomyoma
Skeletal muscle	Rhabdomyoma

Nomenclature Of Benign Neoplasms:

The nomenclature of **benign epithelial tumors** is more complex.
They are classified in different ways:

a. Cells of origin
b. Microscopic pattern
c. Macroscopic pattern

Nomenclature Benign Of Neoplasms:

Adenoma: Is applied to benign epithelial neoplasms producing glandular patterns and to neoplasms derived from glands.

Papilloma: Benign epithelial neoplasm that produces microscopically or macroscopically visible finger-like, exophytic projections from epithelial surfaces.

Cystadenoma: It is a tumor forming large cystic masses.

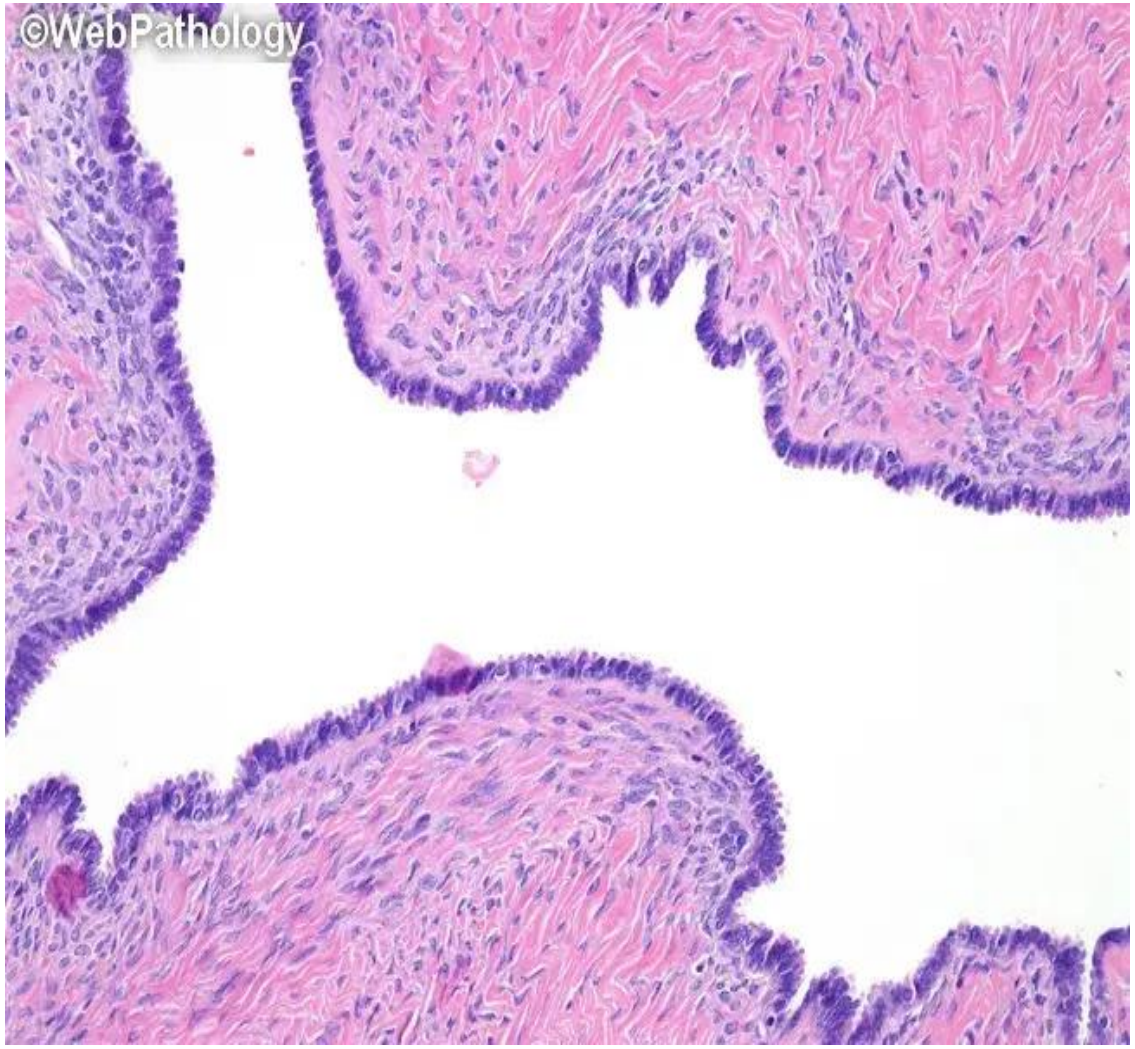
Squamous Papilloma





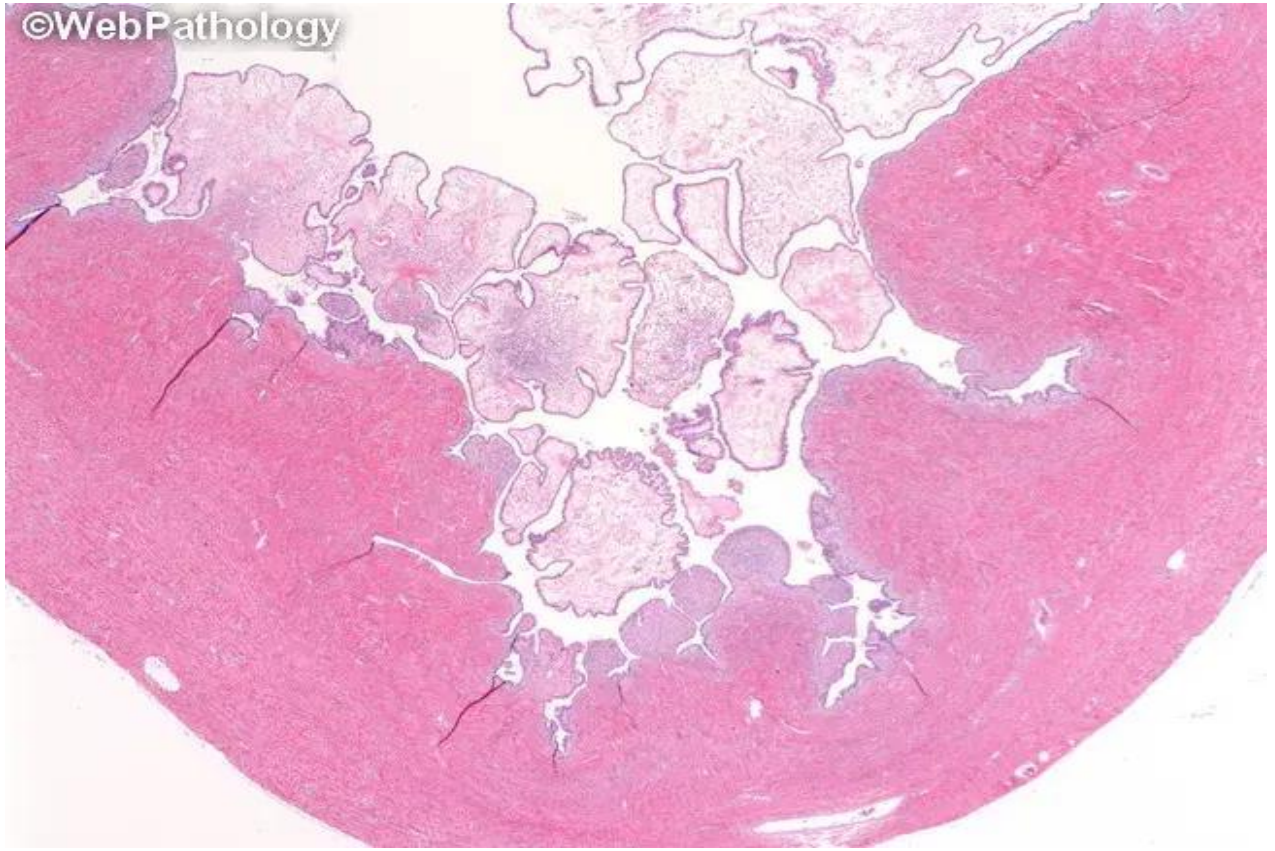
Serous cystadenoma





Papillary serous cystadenoma

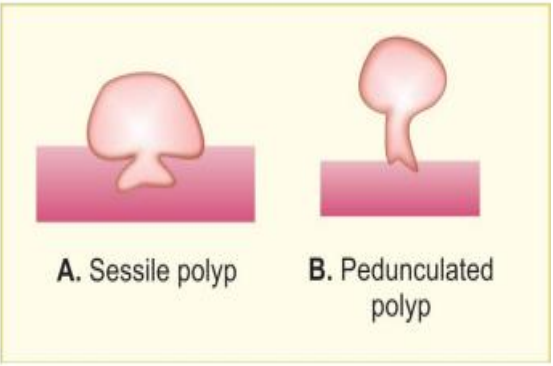




Papillary serous cystadenoma of the ovary**.** The tumor is **multicystic** and shows **fine papillary projections** from the cyst walls focally. The papillary projections, when present, are **lined by a single layer of cuboidal or columnar cells**

Nomenclature Of Benign Neoplasms:

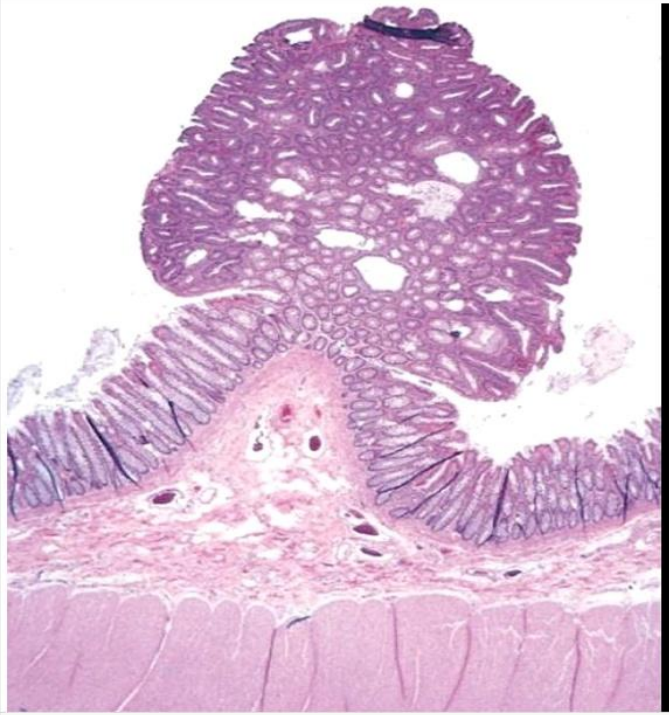
Polyp: It is a projection above a mucosal surface and it projects into the lumen.
It may have a stalk (pedunculated polyp) or may be without a stalk (sessile polyp).



Polyps are broadly divided into:
1. **Non neoplastic:** ex: Inflammatory polyps, hyperplastic polyps.
2. **Neoplastic polyps:** ex: Adenomatous polyps that are lined by dysplastic epithelium.

Polyp

Polyp: A mucosal projection supported by a fibrovascular core





Benign epithelial tumor names

Term	Short definition	Example
Adenoma	Benign epithelial tumor forming glands or arising from glands	Follicular adenoma of thyroid
Papilloma	Benign epithelial tumor with finger-like exophytic projections	Squamous papilloma
Cystadenoma	Benign epithelial tumor forming large cystic masses	Ovarian cystadenoma
Papillary cystadenoma	Papillary structures project into cystic spaces	Papillary serous cystadenoma
Polyp	Projection above a mucosal surface into a lumen	Sessile or pedunculated polyp

Nomenclature Of Malignant Neoplasms:

They are termed as carcinoma or sarcoma depending on the parenchymal cell of origin.

1. **Sarcomas:** They are malignant tumors arising in mesenchymal tissue.
2. **Carcinomas:** They are malignant neoplasms arising from epithelial cell.

Carcinomas that grow in a **glandular pattern** are called **adenocarcinomas**, and Those that produce **squamous cells** are called **squamous cell carcinomas**.

Nomenclature Of Malignant Neoplasms:

Malignant tumors

- Carcinoma: malignant epithelial tumor.
- Sarcoma: malignant mesenchymal tumor.
- Adenocarcinoma: gland-forming carcinoma.
- Squamous cell carcinoma: malignant squamous differentiation.

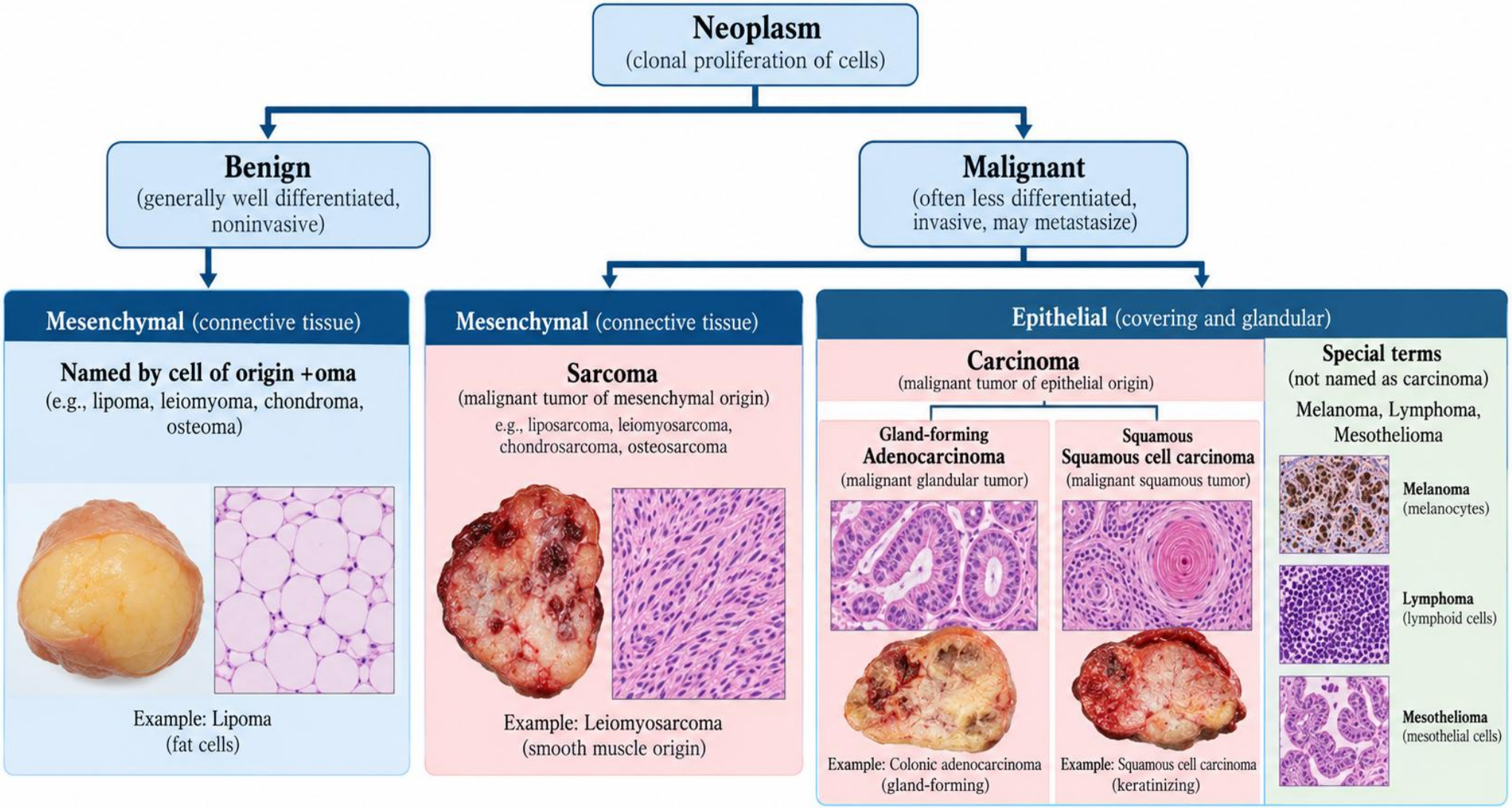
Malignant tumors with “-oma”

- Melanoma.
- Lymphoma.
- Mesothelioma.
- Seminoma / dysgerminoma.
- Hepatoma is often used for hepatocellular carcinoma.

“Omas” that are not true neoplasms

- Hamartoma: disorganized overgrowth of tissue in normal location.
- Choristoma: normal tissue in an abnormal location.
- Granuloma: chronic inflammatory lesion, not a neoplasm.

Tumor nomenclature:



Nomenclature Of Malignant Neoplasms:

TABLE 7.1: Nomenclature of few benign and malignant mesenchymal tumors

Cell of origin	*Benign*	*Malignant*
Fibrous tissue	Fibroma	Fibrosarcoma
Fat cell	Lipoma	Liposarcoma
Blood vessel	Hemangioma	Angiosarcoma
Cartilage	Chondroma	Chondrosarcoma
Bone	Osteoma	Osteogenic sarcoma
Smooth muscle	Leiomyoma	Leiomyosarcoma
Skeletal muscle	Rhabdomyoma	Rhabdomyosarcoma

Summary

- Neoplasm = clonal abnormal growth that persists beyond stimulus.
- Parenchyma determines nomenclature and biologic behavior; stroma supports growth.
- Benign tumors are usually localized and well differentiated.
- Malignant tumors invade and may metastasize.
- Naming depends on tissue of origin, growth pattern and behavior.

Thank you